PHỤ LỤC B — CẤU TRÚC BẢNG VÀ RÀNG BUỘC

TASTORIA

Bản biên tập ngày 31/10/2026

MỤC LỤC

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CHƯƠNG 1

PHỤ LỤC B — CẤU TRÚC BẢNG VÀ RÀNG BUỘC

Nguồn: `Tastoria/db/schema.sql`, SHA-256 `d626650dcfb6c604d4707ea16b8af486e1ef9575cfd6bbe5c6e9b2d219638252`. Có 62 bảng trong tệp nguồn. Các migration bổ sung phải đọc cùng khi mô tả DB đã triển khai; phụ lục này không giả định đã truy vấn schema production. DB đã triển khai có 63 bảng: migration 0003 thêm `recipe_make_ahead` (món làm sẵn; cột và ràng buộc ở `docs/sad/05-tu-dien-du-lieu.md`); migration 0002 chỉ thêm dữ liệu tham chiếu.

1.1. idempotency_keys

Bảng 1.1. idempotency_keys

Cột	Kiểu và ràng buộc khai báo
user_id	uuid NOT NULL
key	text NOT NULL
request_sha256	bytea NOT NULL
status_code	smallint NOT NULL
response_body	jsonb NOT NULL
created_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_idempotency_keys PRIMARY KEY (user_id, key)

1.2. users

Bảng 1.2. users

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
email	citext NOT NULL
password_hash	text NOT NULL

Bảng 1.2 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo
role	user_role_t NOT NULL DEFAULT 'user'
status	user_status_t NOT NULL DEFAULT 'active'
token_version	integer NOT NULL DEFAULT 0
failed_logins	smallint NOT NULL DEFAULT 0
locked_until	timestamptz
created_at	timestamptz NOT NULL DEFAULT now()
updated_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_users PRIMARY KEY (id)
- CONSTRAINT uq_users_email UNIQUE (email)
- CONSTRAINT ck_users_token_version CHECK (token_version >= 0)
- CONSTRAINT ck_users_failed_logins CHECK (failed_logins >= 0)

1.3. refresh_tokens

Bảng 1.3. refresh_tokens

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
user_id	uuid NOT NULL
family_id	uuid NOT NULL
token_sha256	bytea NOT NULL
issued_at	timestamptz NOT NULL
expires_at	timestamptz NOT NULL
revoked_at	timestamptz
replaced_by_id	uuid
user_agent	text

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_refresh_tokens PRIMARY KEY (id)

- CONSTRAINT uq_refresh_tokens_token_sha256 UNIQUE (token_sha256)
- CONSTRAINT uq_refresh_tokens_replaced_by_id UNIQUE (replaced_by_id)
- CONSTRAINT ck_refresh_tokens_expiry CHECK (expires_at > issued_at)
- CONSTRAINT ck_refresh_tokens_not_self CHECK (replaced_by_id <> id)
- CONSTRAINT fk_refresh_tokens_user_id FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE CASCADE
- CONSTRAINT fk_refresh_tokens_replaced_by_id FOREIGN KEY (replaced_by_id) REFERENCES refresh_tokens (id)

1.4. password_resets

Bảng 1.4. password_resets

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
user_id	uuid NOT NULL
token_sha256	bytea NOT NULL
created_at	timestamptz NOT NULL DEFAULT now()
expires_at	timestamptz NOT NULL
used_at	timestamptz

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_password_resets PRIMARY KEY (id)
- CONSTRAINT uq_password_resets_token_sha256 UNIQUE (token_sha256)
- CONSTRAINT ck_password_resets_expiry CHECK (expires_at > created_at)
- CONSTRAINT fk_password_resets_user_id FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE CASCADE

1.5. consent_purposes

Bảng 1.5. consent_purposes

Cột	Kiểu và ràng buộc khai báo
code	text NOT NULL

Bảng 1.5 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo
name_vi	text NOT NULL
description_vi	text NOT NULL
data_category	text NOT NULL
is_sensitive	boolean NOT NULL
is_required	boolean NOT NULL DEFAULT false

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_consent_purposes PRIMARY KEY (code)

1.6. policy_versions

Bảng 1.6. policy_versions

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
purpose_code	text NOT NULL
version	text NOT NULL
text_sha256	bytea NOT NULL
effective_from	timestamptz NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_policy_versions PRIMARY KEY (id)
- CONSTRAINT uq_policy_versions_purpose_code_version UNIQUE (purpose_code, version)
- CONSTRAINT fk_policy_versions_purpose_code FOREIGN KEY (purpose_code) REFERENCES consent_purposes (code)

1.7. consent_events

Bảng 1.7. consent_events

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
user_id	uuid NOT NULL
policy_version_id	uuid NOT NULL
action	consent_action_t NOT NULL
channel	consent_channel_t NOT NULL
app_version	text
created_at	timestamptz NOT NULL DEFAULT clock_timestamp()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_consent_events PRIMARY KEY (id)
- CONSTRAINT fk_consent_events_user_id FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE CASCADE
- CONSTRAINT fk_consent_events_policy_version_id FOREIGN KEY (policy_version_id) REFERENCES policy_versions (id)

1.8. dsr_requests

Bảng 1.8. dsr_requests

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
user_id	uuid
subject_ref	bytea NOT NULL
type	dsr_type_t NOT NULL
status	dsr_status_t NOT NULL
received_at	timestamptz NOT NULL
acknowledged_at	timestamptz
completed_at	timestamptz
result_key	text
note	text

Bảng 1.8 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo

Ràng buộc đặt tên trong CREATE TABLE:

- `CONSTRAINT pk_dsr_requests PRIMARY KEY (id)`
- `CONSTRAINT ck_dsr_requests_completed_at CHECK (completed_at IS NULL OR completed_at >= received_at)`
- `CONSTRAINT fk_dsr_requests_user_id FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE SET NULL`

1.9. audit_log

Bảng 1.9. audit_log

Cột	Kiểu và ràng buộc khai báo
`id`	`bigint GENERATED ALWAYS AS IDENTITY`
`actor_user_id`	`uuid`
`actor_role`	`text NOT NULL`
`action`	`text NOT NULL`
`entity`	`text NOT NULL`
`entity_id`	`text`
`purpose`	`text`
`request_id`	`text`
`created_at`	`timestamptz NOT NULL DEFAULT now()`

Ràng buộc đặt tên trong CREATE TABLE:

- `CONSTRAINT pk_audit_log PRIMARY KEY (id)`
- `CONSTRAINT fk_audit_log_actor_user_id FOREIGN KEY (actor_user_id) REFERENCES users (id) ON DELETE SET NULL`

1.10. food_groups

Bảng 1.10. food_groups

Cột	Kiểu và ràng buộc khai báo
code	text NOT NULL
name_vi	text NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_food_groups PRIMARY KEY (code)

1.11. nutrient_sources

Bảng 1.11. nutrient_sources

Cột	Kiểu và ràng buộc khai báo
id	smallint NOT NULL
code	text NOT NULL
name	text NOT NULL
version	text NOT NULL
license	text NOT NULL
url	text NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_nutrient_sources PRIMARY KEY (id)
- CONSTRAINT uq_nutrient_sources_code UNIQUE (code)

1.12. food_items

Bảng 1.12. food_items

Cột	Kiểu và ràng buộc khai báo
source_id	smallint NOT NULL
source_code	text NOT NULL
name	text NOT NULL
name_en	text

Bảng 1.12 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo
refuse_pct	numeric

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_food_items PRIMARY KEY (source_id, source_code)
- CONSTRAINT ck_food_items_refuse_pct CHECK (refuse_pct BETWEEN 0 AND 100)
- CONSTRAINT fk_food_items_source_id FOREIGN KEY (source_id) REFERENCES nutrient_sources (id)

1.13. nutrients

Bảng 1.13. nutrients

Cột	Kiểu và ràng buộc khai báo
code	text NOT NULL
name_vi	text NOT NULL
unit	text NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_nutrients PRIMARY KEY (code)

1.14. food_nutrients

Bảng 1.14. food_nutrients

Cột	Kiểu và ràng buộc khai báo
source_id	smallint NOT NULL
source_code	text NOT NULL
nutrient_code	text NOT NULL
per_100g	numeric NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_food_nutrients PRIMARY KEY (source_id, source_code, nutrient_code)
- CONSTRAINT ck_food_nutrients_per_100g CHECK (per_100g >= 0)
- CONSTRAINT fk_food_nutrients_nutrient_code FOREIGN KEY (nutrient_code) REFERENCES nutrients (code)
- CONSTRAINT fk_food_nutrients_source_id_source_code FOREIGN KEY (source_id, source_code) REFERENCES food_items (source_id, source_code)

1.15. allergen_groups

Bảng 1.15. allergen_groups

Cột	Kiểu và ràng buộc khai báo
code	text NOT NULL
name_vi	text NOT NULL
name_en	text NOT NULL
basis	allergen_basis_t NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_allergen_groups PRIMARY KEY (code)

1.16. diets

Bảng 1.16. diets

Cột	Kiểu và ràng buộc khai báo
code	text NOT NULL
name_vi	text NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_diets PRIMARY KEY (code)

1.17. diet_exclusions

Bảng 1.17. diet_exclusions

Cột	Kiểu và ràng buộc khai báo
diet_code	text NOT NULL
food_group_code	text NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_diet_exclusions PRIMARY KEY (diet_code, food_group_code)
- CONSTRAINT fk_diet_exclusions_diet_code FOREIGN KEY (diet_code) REFERENCES diets (code)
- CONSTRAINT fk_diet_exclusions_food_group_code FOREIGN KEY (food_group_code) REFERENCES food_groups (code)

1.18. ingredients

Bảng 1.18. ingredients

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
name_vi	text NOT NULL
food_group_code	text NOT NULL
comp_source_id	smallint
comp_source_code	text
is_staple	boolean NOT NULL DEFAULT false
countable	boolean NOT NULL DEFAULT false
shelf_life_days	numeric
shelf_life_basis	text
is_active	boolean NOT NULL DEFAULT true
created_at	timestamptz NOT NULL DEFAULT now()
updated_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_ingredients PRIMARY KEY (id)
- CONSTRAINT ck_ingredients_comp_pair CHECK ((comp_source_id IS NULL) = (comp_source_code IS NULL))

- CONSTRAINT ck_ingredients_shelf_life_days CHECK (shelf_life_days > 0)
- CONSTRAINT fk_ingredients_food_group_code FOREIGN KEY (food_group_code) REFERENCES food_groups (code)
- CONSTRAINT fk_ingredients_comp_source_id_comp_source_code FOREIGN KEY (comp_source_id, comp_source_code) REFERENCES food_items (source_id, source_code)

1.19. ingredient_synonyms

Bảng 1.19. ingredient_synonyms

Cột	Kiểu và ràng buộc khai báo
ingredient_id	uuid NOT NULL
synonym	text NOT NULL
lang	lang_t NOT NULL
source	text

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_ingredient_synonyms PRIMARY KEY (ingredient_id, synonym, lang)
- CONSTRAINT fk_ingredient_synonyms_ingredient_id FOREIGN KEY (ingredient_id) REFERENCES ingredients (id)

1.20. ingredient_allergens

Bảng 1.20. ingredient_allergens

Cột	Kiểu và ràng buộc khai báo
ingredient_id	uuid NOT NULL
allergen_code	text NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_ingredient_allergens PRIMARY KEY (ingredient_id, allergen_code)
- CONSTRAINT fk_ingredient_allergens_ingredient_id FOREIGN KEY (ingredient_id) REFERENCES ingredients (id)

- CONSTRAINT fk_ingredient_allergens_allergen_code FOREIGN KEY (allergen_code) REFERENCES allergen_groups (code)

1.21. ingredient_units

Bảng 1.21. ingredient_units

Cột	Kiểu và ràng buộc khai báo
ingredient_id	uuid NOT NULL
unit_code	text NOT NULL
grams_per_unit	numeric NOT NULL
source	text NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_ingredient_units PRIMARY KEY (ingredient_id, unit_code)
- CONSTRAINT ck_ingredient_units_grams_per_unit CHECK (grams_per_unit > 0)
- CONSTRAINT fk_ingredient_units_ingredient_id FOREIGN KEY (ingredient_id) REFERENCES ingredients (id)

1.22. qty_levels

Bảng 1.22. qty_levels

Cột	Kiểu và ràng buộc khai báo
food_group_code	text NOT NULL
level	qty_level_t NOT NULL
g_min	numeric NOT NULL
g_max	numeric NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_qty_levels PRIMARY KEY (food_group_code, level)
- CONSTRAINT ck_qty_levels_1 CHECK (g_min <= g_max)
- CONSTRAINT ck_qty_levels_g_min CHECK (g_min >= 0)
- CONSTRAINT fk_qty_levels_food_group_code FOREIGN KEY (food_group_code) REFERENCES food_groups (code)

1.23. cooking_methods

Bảng 1.23. cooking_methods

Cột	Kiểu và ràng buộc khai báo
code	text NOT NULL
name_vi	text NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_cooking_methods PRIMARY KEY (code)

1.24. tags

Bảng 1.24. tags

Cột	Kiểu và ràng buộc khai báo
code	text NOT NULL
kind	tag_kind_t NOT NULL
name_vi	text NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_tags PRIMARY KEY (code)

1.25. recipes

Bảng 1.25. recipes

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
name_vi	text NOT NULL
slot	slot_t NOT NULL
servings	smallint NOT NULL
method_code	text NOT NULL

Bảng 1.25 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo
status	recipe_status_t NOT NULL DEFAULT 'draft'
notes	text
created_at	timestamptz NOT NULL DEFAULT now()
updated_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_recipes PRIMARY KEY (id)
- CONSTRAINT uq_recipes_id_slot UNIQUE (id, slot)
- CONSTRAINT ck_recipes_servings CHECK (servings > 0)
- CONSTRAINT fk_recipes_method_code FOREIGN KEY (method_code) REFERENCES cooking_methods (code)

1.26. recipe_ingredients

Bảng 1.26. recipe_ingredients

Cột	Kiểu và ràng buộc khai báo
recipe_id	uuid NOT NULL
ingredient_id	uuid NOT NULL
grams	numeric NOT NULL
role	ingredient_role_t NOT NULL
is_optional	boolean NOT NULL DEFAULT false

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_recipe_ingredients PRIMARY KEY (recipe_id, ingredient_id)
- CONSTRAINT ck_recipe_ingredients_grams CHECK (grams > 0)
- CONSTRAINT fk_recipe_ingredients_recipe_id FOREIGN KEY (recipe_id) REFERENCES recipes (id)
- CONSTRAINT fk_recipe_ingredients_ingredient_id FOREIGN KEY (ingredient_id) REFERENCES ingredients (id)

1.27. recipe_steps

Bảng 1.27. recipe_steps

Cột	Kiểu và ràng buộc khai báo
recipe_id	uuid NOT NULL
step_no	smallint NOT NULL
text_vi	text NOT NULL
kind	step_kind_t NOT NULL
minutes	smallint NOT NULL
uses_burner	boolean NOT NULL DEFAULT false

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_recipe_steps PRIMARY KEY (recipe_id, step_no)
- CONSTRAINT ck_recipe_steps_minutes CHECK (minutes > 0)
- CONSTRAINT ck_recipe_steps_step_no CHECK (step_no > 0)
- CONSTRAINT fk_recipe_steps_recipe_id FOREIGN KEY (recipe_id) REFERENCES recipes (id)

1.28. recipe_tags

Bảng 1.28. recipe_tags

Cột	Kiểu và ràng buộc khai báo
recipe_id	uuid NOT NULL
tag_code	text NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_recipe_tags PRIMARY KEY (recipe_id, tag_code)
- CONSTRAINT fk_recipe_tags_recipe_id FOREIGN KEY (recipe_id) REFERENCES recipes (id)
- CONSTRAINT fk_recipe_tags_tag_code FOREIGN KEY (tag_code) REFERENCES tags (code)

1.29. recipe_sources

Bảng 1.29. recipe_sources

Cột	Kiểu và ràng buộc khai báo
recipe_id	uuid NOT NULL
url	text NOT NULL
title	text NOT NULL
accessed_on	date NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_recipe_sources PRIMARY KEY (recipe_id, url)
- CONSTRAINT fk_recipe_sources_recipe_id FOREIGN KEY (recipe_id) REFERENCES recipes (id)

1.30. recipe_nutrition

Bảng 1.30. recipe_nutrition

Cột	Kiểu và ràng buộc khai báo
recipe_id	uuid NOT NULL
nutrient_code	text NOT NULL
per_serving	numeric NOT NULL
computed_at	timestamptz NOT NULL
data_version	text NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_recipe_nutrition PRIMARY KEY (recipe_id, nutrient_code)
- CONSTRAINT ck_recipe_nutrition_per_serving CHECK (per_serving >= 0)
- CONSTRAINT fk_recipe_nutrition_recipe_id FOREIGN KEY (recipe_id) REFERENCES recipes (id)
- CONSTRAINT fk_recipe_nutrition_nutrient_code FOREIGN KEY (nutrient_code) REFERENCES nutrients (code)

1.31. vision_classes

Bảng 1.31. vision_classes

Cột	Kiểu và ràng buộc khai báo
id	smallint NOT NULL
code	text NOT NULL
name_vi	text NOT NULL
name_en	text NOT NULL
countable	boolean NOT NULL
instance_def	instance_def_t NOT NULL
has_freshness_model	boolean NOT NULL DEFAULT false

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_vision_classes PRIMARY KEY (id)
- CONSTRAINT uq_vision_classes_code UNIQUE (code)

1.32. vision_class_ingredients

Bảng 1.32. vision_class_ingredients

Cột	Kiểu và ràng buộc khai báo
vision_class_id	smallint NOT NULL
ingredient_id	uuid NOT NULL
is_default	boolean NOT NULL DEFAULT false

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_vision_class_ingredients PRIMARY KEY (vision_class_id, ingredient_id)
- CONSTRAINT fk_vision_class_ingredients_vision_class_id FOREIGN KEY (vision_class_id) REFERENCES vision_classes (id)
- CONSTRAINT fk_vision_class_ingredients_ingredient_id FOREIGN KEY (ingredient_id) REFERENCES ingredients (id)

1.33. profiles

Bảng 1.33. profiles

Cột	Kiểu và ràng buộc khai báo
user_id	uuid NOT NULL
sex_enc	bytea
birth_year_enc	bytea
height_cm_enc	bytea
weight_kg_enc	bytea
enc_key_id	text
activity_level	activity_level_t
goal	goal_t
kcal_target	integer
kcal_source	kcal_source_t
diet_code	text
updated_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_profiles PRIMARY KEY (user_id)
- CONSTRAINT ck_profiles_kcal_target CHECK (kcal_target BETWEEN 800 AND 5000)
- CONSTRAINT ck_profiles_kcal_pair CHECK ((kcal_target IS NULL) = (kcal_source IS NULL))
- CONSTRAINT ck_profiles_expert CHECK (kcal_source IS NULL OR (goal IS NOT DISTINCT FROM 'expert') = (kcal_source = 'expert'))
- CONSTRAINT ck_profiles_enc_key CHECK (num_nonnulls(sex_enc, birth_year_enc, height_cm_enc, weight_kg_enc) = 0 OR enc_key_id IS NOT NULL)
- CONSTRAINT fk_profiles_user_id FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE CASCADE
- CONSTRAINT fk_profiles_diet_code FOREIGN KEY (diet_code) REFERENCES diets (code)

1.34. user_allergens

Bảng 1.34. user_allergens

Cột	Kiểu và ràng buộc khai báo
user_id	uuid NOT NULL
allergen_code	text NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_user_allergens PRIMARY KEY (user_id, allergen_code)
- CONSTRAINT fk_user_allergens_user_id FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE CASCADE
- CONSTRAINT fk_user_allergens_allergen_code FOREIGN KEY (allergen_code) REFERENCES allergen_groups (code)

1.35. user_favorite_recipes

Bảng 1.35. user_favorite_recipes

Cột	Kiểu và ràng buộc khai báo
user_id	uuid NOT NULL
recipe_id	uuid NOT NULL
created_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_user_favorite_recipes PRIMARY KEY (user_id, recipe_id)
- CONSTRAINT fk_user_favorite_recipes_user_id FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE CASCADE
- CONSTRAINT fk_user_favorite_recipes_recipe_id FOREIGN KEY (recipe_id) REFERENCES recipes (id)

1.36. user_settings

Bảng 1.36. user_settings

Cột	Kiểu và ràng buộc khai báo
user_id	uuid NOT NULL

Bảng 1.36 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo
burners	smallint NOT NULL DEFAULT 2
default_persons	smallint NOT NULL DEFAULT 2
timezone	text NOT NULL DEFAULT 'Asia/Ho_Chi_Minh'
updated_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_user_settings PRIMARY KEY (user_id)
- CONSTRAINT ck_user_settings_burners CHECK (burners BETWEEN 1 AND 4)
- CONSTRAINT ck_user_settings_default_persons CHECK (default_persons BETWEEN 1 AND 6)
- CONSTRAINT fk_user_settings_user_id FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE CASCADE

1.37. pantry_lots

Bảng 1.37. pantry_lots

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
user_id	uuid NOT NULL
ingredient_id	uuid NOT NULL
initial_g_min	numeric NOT NULL
initial_g_max	numeric NOT NULL
acquired_at	timestamptz NOT NULL
acquired_basis	acquired_basis_t NOT NULL
status	lot_status_t NOT NULL DEFAULT 'in_stock'
p_tuoi	real
p_dung_som	real
p_dung_ngay	real
p_kiem_tra	real

Bảng 1.37 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo
freshness_observed_at	timestamptz
freshness_model_version_id	uuid
user_checked_at	timestamptz
source_scan_item_id	uuid
created_at	timestamptz NOT NULL DEFAULT now()
updated_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_pantry_lots PRIMARY KEY (id)
- CONSTRAINT ck_pantry_lots_initial_g_min CHECK (initial_g_min >= 0)
- CONSTRAINT ck_pantry_lots_1 CHECK (initial_g_min <= initial_g_max)
- CONSTRAINT ck_pantry_lots_p_range CHECK (p_tuoi BETWEEN 0 AND 1 AND p_dung_som BETWEEN 0 AND 1 AND p_dung_ngay BETWEEN 0 AND 1 AND p_kiem_tra BETWEEN 0 AND 1)
- CONSTRAINT ck_pantry_lots_p_group CHECK (num_nulls(p_tuoi, p_dung_som, p_dung_ngay, p_kiem_tra) IN (0, 4))
- CONSTRAINT ck_pantry_lots_p_sum CHECK (p_tuoi IS NULL OR abs(p_tuoi + p_dung_som + p_dung_ngay + p_kiem_tra - 1) <= 0.01)
- CONSTRAINT ck_pantry_lots_p_observed CHECK (p_tuoi IS NULL OR freshness_observed_at IS NOT NULL)
- CONSTRAINT fk_pantry_lots_user_id FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE CASCADE
- CONSTRAINT fk_pantry_lots_ingredient_id FOREIGN KEY (ingredient_id) REFERENCES ingredients (id)

1.38. pantry_transactions

Bảng 1.38. pantry_transactions

Cột	Kiểu và ràng buộc khai báo
id	bigint GENERATED ALWAYS AS IDENTITY
lot_id	uuid NOT NULL
delta_g_min	numeric NOT NULL

Bảng 1.38 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo
delta_g_max	numeric NOT NULL
reason	pantry_tx_reason_t NOT NULL
ref_type	pantry_ref_type_t
ref_id	uuid
reverses_id	bigint
urgency_at_event	real
created_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_pantry_transactions PRIMARY KEY (id)
- CONSTRAINT uq_pantry_transactions_reverses_id UNIQUE (reverses_id)
- CONSTRAINT ck_pantry_transactions_ref_pair CHECK ((ref_type IS NULL) = (ref_id IS NULL))
- CONSTRAINT ck_pantry_transactions_undo CHECK ((reason = 'undo') = (reverses_id IS NOT NULL))
- CONSTRAINT ck_pantry_transactions_not_self CHECK (reverses_id <> id)
- CONSTRAINT ck_pantry_transactions_urgency CHECK (urgency_at_event >= 0)
- CONSTRAINT fk_pantry_transactions_lot_id FOREIGN KEY (lot_id) REFERENCES pantry_lots (id) ON DELETE CASCADE
- CONSTRAINT fk_pantry_transactions_reverses_id FOREIGN KEY (reverses_id) REFERENCES pantry_transactions (id)

1.39. model_versions

Bảng 1.39. model_versions

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
name	model_name_t NOT NULL
version	text NOT NULL
artifact_key	text NOT NULL
sha256	bytea NOT NULL

Bảng 1.39 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo
preprocess_version	text
metrics	jsonb NOT NULL
gate_report_key	text
mlflow_run_id	text
status	model_status_t NOT NULL DEFAULT 'candidate'
created_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_model_versions PRIMARY KEY (id)
- CONSTRAINT uq_model_versions_name_version UNIQUE (name, version)
- CONSTRAINT uq_model_versions_id_name UNIQUE (id, name)

1.40. model_version_classes

Bảng 1.40. model_version_classes

Cột	Kiểu và ràng buộc khai báo
model_version_id	uuid NOT NULL
class_index	smallint NOT NULL
vision_class_id	smallint NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_model_version_classes PRIMARY KEY (model_version_id, class_index)
- CONSTRAINT uq_model_version_classes_model_version_id_vision_class_id UNIQUE (model_version_id, vision_class_id)
- CONSTRAINT ck_model_version_classes_class_index CHECK (class_index >= 0)
- CONSTRAINT fk_model_version_classes_model_version_id FOREIGN KEY (model_version_id) REFERENCES model_versions (id)
- CONSTRAINT fk_model_version_classes_vision_class_id FOREIGN KEY (vision_class_id) REFERENCES vision_classes (id)

1.41. model_deployments

Bảng 1.41. model_deployments

Cột	Kiểu và ràng buộc khai báo
name	model_name_t NOT NULL
active_version_id	uuid NOT NULL
previous_version_id	uuid
switched_at	timestamptz NOT NULL
switched_by	uuid

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_model_deployments PRIMARY KEY (name)
- CONSTRAINT uq_model_deployments_active_version_id UNIQUE (active_version_id)
- CONSTRAINT ck_model_deployments_previous CHECK (previous_version_id <> active_version_id)
- CONSTRAINT fk_model_deployments_active_version_id_name FOREIGN KEY (active_version_id, name) REFERENCES model_versions (id, name)
- CONSTRAINT fk_model_deployments_previous_version_id_name FOREIGN KEY (previous_version_id, name) REFERENCES model_versions (id, name)
- CONSTRAINT fk_model_deployments_switched_by FOREIGN KEY (switched_by) REFERENCES users (id) ON DELETE SET NULL

1.42. scans

Bảng 1.42. scans

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
user_id	uuid NOT NULL
mode	scan_mode_t NOT NULL
status	scan_status_t NOT NULL
error_code	text

Bảng 1.42 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo
attempts	smallint NOT NULL DEFAULT 0
detector_version_id	uuid
classifier_version_id	uuid
preprocess_version	text
enqueued_at	timestamptz
started_at	timestamptz
finished_at	timestamptz
confirmed_at	timestamptz
created_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_scans PRIMARY KEY (id)
- CONSTRAINT ck_scans_attempts CHECK (attempts >= 0)
- CONSTRAINT ck_scans_confirmed CHECK ((status = 'confirmed') = (confirmed_at IS NOT NULL))
- CONSTRAINT fk_scans_user_id FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE CASCADE
- CONSTRAINT fk_scans_detector_version_id FOREIGN KEY (detector_version_id) REFERENCES model_versions (id)
- CONSTRAINT fk_scans_classifier_version_id FOREIGN KEY (classifier_version_id) REFERENCES model_versions (id)

1.43. scan_images

Bảng 1.43. scan_images

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
scan_id	uuid NOT NULL
ordinal	smallint NOT NULL
shelf_group	smallint NOT NULL DEFAULT 1
storage_key	text NOT NULL

Bảng 1.43 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo
width	integer NOT NULL
height	integer NOT NULL
contributed	boolean NOT NULL DEFAULT false
purged_at	timestamptz

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_scan_images PRIMARY KEY (id)
- CONSTRAINT uq_scan_images_storage_key UNIQUE (storage_key)
- CONSTRAINT uq_scan_images_scan_id_ordinal UNIQUE (scan_id, ordinal)
- CONSTRAINT ck_scan_images_ordinal CHECK (ordinal BETWEEN 1 AND 2)
- CONSTRAINT ck_scan_images_size CHECK (width > 0 AND height > 0)
- CONSTRAINT ck_scan_images_shelf_group CHECK (shelf_group >= 1)
- CONSTRAINT fk_scan_images_scan_id FOREIGN KEY (scan_id) REFERENCES scans (id) ON DELETE CASCADE

1.44. scan_items

Bảng 1.44. scan_items

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
scan_id	uuid NOT NULL
ingredient_id	uuid NOT NULL
origin	scan_item_origin_t NOT NULL
model_count	smallint
final_count	smallint
qty_level	qty_level_t
user_action	scan_user_action_t NOT NULL DEFAULT 'kept'
original_ingredient_id	uuid
user_freshness	freshness_t
created_at	timestamptz NOT NULL DEFAULT now()

Bảng 1.44 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo
updated_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_scan_items PRIMARY KEY (id)
- CONSTRAINT ck_scan_items_model_count CHECK (model_count >= 0)
- CONSTRAINT ck_scan_items_final_count CHECK (final_count BETWEEN 0 AND 99)
- CONSTRAINT ck_scan_items_relabeled CHECK (user_action <> 'relabeled' OR original_ingredient_id IS NOT NULL)
- CONSTRAINT ck_scan_items_original CHECK (original_ingredient_id IS NULL OR (origin = 'model' AND original_ingredient_id <> ingredient_id))
- CONSTRAINT ck_scan_items_user_added CHECK (origin = 'model' OR (model_count IS NULL AND user_action <> 'deleted'))
- CONSTRAINT fk_scan_items_scan_id FOREIGN KEY (scan_id) REFERENCES scans (id) ON DELETE CASCADE
- CONSTRAINT fk_scan_items_ingredient_id FOREIGN KEY (ingredient_id) REFERENCES ingredients (id)
- CONSTRAINT fk_scan_items_original_ingredient_id FOREIGN KEY (original_ingredient_id) REFERENCES ingredients (id)

1.45. detections

Bảng 1.45. detections

Cột	Kiểu và ràng buộc khai báo
id	bigint GENERATED ALWAYS AS IDENTITY
scan_image_id	uuid NOT NULL
scan_item_id	uuid
vision_class_id	smallint NOT NULL
x1	real NOT NULL
y1	real NOT NULL
x2	real NOT NULL
y2	real NOT NULL
confidence	real NOT NULL

Bảng 1.45 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo
p_tuoi	real
p_dung_som	real
p_dung_ngay	real
p_kiem_tra	real

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_detections PRIMARY KEY (id)
- CONSTRAINT ck_detections_1 CHECK (0 <= x1 AND x1 < x2 AND x2 <= 1 AND 0 <= y1 AND y1 < y2 AND y2 <= 1)
- CONSTRAINT ck_detections_confidence CHECK (confidence BETWEEN 0 AND 1)
- CONSTRAINT ck_detections_p_range CHECK (p_tuoi BETWEEN 0 AND 1 AND p_dung_som BETWEEN 0 AND 1 AND p_dung_ngay BETWEEN 0 AND 1 AND p_kiem_tra BETWEEN 0 AND 1)
- CONSTRAINT ck_detections_p_group CHECK (num_nulls(p_tuoi, p_dung_som, p_dung_ngay, p_kiem_tra) IN (0, 4))
- CONSTRAINT ck_detections_p_sum CHECK (p_tuoi IS NULL OR abs(p_tuoi + p_dung_som + p_dung_ngay + p_kiem_tra - 1) <= 0.01)
- CONSTRAINT fk_detections_scan_image_id FOREIGN KEY (scan_image_id) REFERENCES scan_images (id) ON DELETE CASCADE
- CONSTRAINT fk_detections_scan_item_id FOREIGN KEY (scan_item_id) REFERENCES scan_items (id) ON DELETE SET NULL
- CONSTRAINT fk_detections_vision_class_id FOREIGN KEY (vision_class_id) REFERENCES vision_classes (id)

1.46. substitution_pairs

Bảng 1.46. substitution_pairs

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
ingredient_id	uuid NOT NULL
substitute_id	uuid NOT NULL
context_slot	slot_t

Bảng 1.46 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo
context_method_code	text
ratio	numeric NOT NULL DEFAULT 1
score	real NOT NULL
source	text NOT NULL
curated	boolean NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_substitution_pairs PRIMARY KEY (id)
- CONSTRAINT uq_substitution_pairs_ctx UNIQUE NULLS NOT DISTINCT (ingredient_id, substitute_id, context_slot, context_method_code)
- CONSTRAINT ck_substitution_pairs_1 CHECK (ingredient_id <> substitute_id)
- CONSTRAINT ck_substitution_pairs_ratio CHECK (ratio > 0)
- CONSTRAINT fk_substitution_pairs_ingredient_id FOREIGN KEY (ingredient_id) REFERENCES ingredients (id)
- CONSTRAINT fk_substitution_pairs_substitute_id FOREIGN KEY (substitute_id) REFERENCES ingredients (id)
- CONSTRAINT fk_substitution_pairs_context_method_code FOREIGN KEY (context_method_code) REFERENCES cooking_methods (code)

1.47. ingredient_embeddings

Bảng 1.47. ingredient_embeddings

Cột	Kiểu và ràng buộc khai báo
ingredient_id	uuid NOT NULL
model_version_id	uuid NOT NULL
vector	vector(100) NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_ingredient_embeddings PRIMARY KEY (ingredient_id, model_version_id)
- CONSTRAINT fk_ingredient_embeddings_ingredient_id FOREIGN KEY (ingredient_id) REFERENCES ingredients (id)

- CONSTRAINT fk_ingredient_embeddings_model_version_id FOREIGN KEY (model_version_id) REFERENCES model_versions (id)

1.48. shopping_items

Bảng 1.48. shopping_items

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
user_id	uuid NOT NULL
ingredient_id	uuid NOT NULL
qty_g	numeric
note	text
status	shopping_status_t NOT NULL DEFAULT 'open'
source_meal_plan_item_id	uuid
created_at	timestamptz NOT NULL DEFAULT now()
updated_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_shopping_items PRIMARY KEY (id)
- CONSTRAINT ck_shopping_items_qty_g CHECK (qty_g > 0)
- CONSTRAINT fk_shopping_items_user_id FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE CASCADE
- CONSTRAINT fk_shopping_items_ingredient_id FOREIGN KEY (ingredient_id) REFERENCES ingredients (id)

1.49. meal_plans

Bảng 1.49. meal_plans

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
user_id	uuid NOT NULL

Bảng 1.49 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo
meal	meal_t NOT NULL
persons	smallint NOT NULL
max_minutes	smallint NOT NULL
status	meal_plan_status_t NOT NULL DEFAULT 'proposed'
e_target_kcal	numeric
kcal_low	numeric
kcal_high	numeric
rice_bowls_per_person	numeric
reason_vi	text NOT NULL
ranker_version_id	uuid
params_version	text NOT NULL
created_at	timestamptz NOT NULL DEFAULT now()
updated_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_meal_plans PRIMARY KEY (id)
- CONSTRAINT uq_meal_plans_id_user_id UNIQUE (id, user_id)
- CONSTRAINT ck_meal_plans_persons CHECK (persons BETWEEN 1 AND 6)
- CONSTRAINT ck_meal_plans_max_minutes CHECK (max_minutes BETWEEN 10 AND 180)
- CONSTRAINT ck_meal_plans_e_target_kcal CHECK (e_target_kcal > 0)
- CONSTRAINT ck_meal_plans_kcal_pair CHECK ((kcal_low IS NULL) = (kcal_high IS NULL))
- CONSTRAINT ck_meal_plans_kcal_range CHECK (kcal_low >= 0 AND kcal_low <= kcal_high)
- CONSTRAINT ck_meal_plans_rice_bowls CHECK (rice_bowls_per_person BETWEEN 0.5 AND 3 AND rice_bowls_per_person * 2 = trunc(rice_bowls_per_person * 2))
- CONSTRAINT fk_meal_plans_user_id FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE CASCADE
- CONSTRAINT fk_meal_plans_ranker_version_id FOREIGN KEY (ranker_version_id) REFERENCES model_versions (id)

1.50. meal_plan_coeater_allergens

Bảng 1.50. meal_plan_coeater_allergens

Cột	Kiểu và ràng buộc khai báo
meal_plan_id	uuid NOT NULL
allergen_code	text NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_meal_plan_coeater_allergens PRIMARY KEY (meal_plan_id, allergen_code)
- CONSTRAINT fk_meal_plan_coeater_allergens_meal_plan_id FOREIGN KEY (meal_plan_id) REFERENCES meal_plans (id) ON DELETE CASCADE
- CONSTRAINT fk_meal_plan_coeater_allergens_allergen_code FOREIGN KEY (allergen_code) REFERENCES allergen_groups (code)

1.51. meal_plan_pins

Bảng 1.51. meal_plan_pins

Cột	Kiểu và ràng buộc khai báo
meal_plan_id	uuid NOT NULL
lot_id	uuid NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_meal_plan_pins PRIMARY KEY (meal_plan_id, lot_id)
- CONSTRAINT fk_meal_plan_pins_meal_plan_id FOREIGN KEY (meal_plan_id) REFERENCES meal_plans (id) ON DELETE CASCADE
- CONSTRAINT fk_meal_plan_pins_lot_id FOREIGN KEY (lot_id) REFERENCES pantry_lots (id) ON DELETE CASCADE

1.52. meal_plan_items

Bảng 1.52. meal_plan_items

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
meal_plan_id	uuid NOT NULL
slot	slot_t NOT NULL
recipe_id	uuid NOT NULL
status	plan_item_status_t NOT NULL DEFAULT 'active'
replaced_item_id	uuid
swap_reason	swap_reason_t
rank_score	real NOT NULL
servings_scale	numeric NOT NULL
kcal_per_person	numeric
created_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_meal_plan_items PRIMARY KEY (id)
- CONSTRAINT uq_meal_plan_items_replaced_item_id UNIQUE (replaced_item_id)
- CONSTRAINT uq_meal_plan_items_id_meal_plan_id_recipe_id UNIQUE (id, meal_plan_id, recipe_id)
- CONSTRAINT ck_meal_plan_items_servings_scale CHECK (servings_scale > 0)
- CONSTRAINT ck_meal_plan_items_kcal_per_person CHECK (kcal_per_person >= 0)
- CONSTRAINT ck_meal_plan_items_not_self CHECK (replaced_item_id <> id)
- CONSTRAINT fk_meal_plan_items_meal_plan_id FOREIGN KEY (meal_plan_id) REFERENCES meal_plans (id) ON DELETE CASCADE
- CONSTRAINT fk_meal_plan_items_replaced_item_id FOREIGN KEY (replaced_item_id) REFERENCES meal_plan_items (id)
- CONSTRAINT fk_meal_plan_items_recipe_id_slot FOREIGN KEY (recipe_id, slot) REFERENCES recipes (id, slot)

1.53. meal_plan_item_substitutions

Bảng 1.53. meal_plan_item_substitutions

Cột	Kiểu và ràng buộc khai báo
item_id	uuid NOT NULL
original_ingredient_id	uuid NOT NULL
substitute_ingredient_id	uuid NOT NULL
grams	numeric NOT NULL
origin	substitution_origin_t NOT NULL
created_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_meal_plan_item_substitutions PRIMARY KEY (item_id, original_ingredient_id)
- CONSTRAINT ck_meal_plan_item_substitutions_grams CHECK (grams > 0)
- CONSTRAINT ck_meal_plan_item_substitutions_distinct CHECK (original_ingredient_id <> substitute_ingredient_id)
- CONSTRAINT fk_meal_plan_item_substitutions_item_id FOREIGN KEY (item_id) REFERENCES meal_plan_items (id) ON DELETE CASCADE
- CONSTRAINT fk_meal_plan_item_substitutions_original_ingredient_id FOREIGN KEY (original_ingredient_id) REFERENCES ingredients (id)
- CONSTRAINT fk_meal_plan_item_substitutions_substitute_ingredient_id FOREIGN KEY (substitute_ingredient_id) REFERENCES ingredients (id)

1.54. cook_sessions

Bảng 1.54. cook_sessions

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
meal_plan_id	uuid NOT NULL
status	cook_status_t NOT NULL DEFAULT 'in_progress'
serve_at	timestamptz NOT NULL
burners	smallint NOT NULL
started_at	timestamptz NOT NULL

Bảng 1.54 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo
completed_at	timestamptz
reverted_at	timestamptz
created_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_cook_sessions PRIMARY KEY (id)
- CONSTRAINT ck_cook_sessions_burners CHECK (burners BETWEEN 1 AND 4)
- CONSTRAINT ck_cook_sessions_completed CHECK ((status = 'completed') = (completed_at IS NOT NULL AND reverted_at IS NULL))
- CONSTRAINT ck_cook_sessions_reverted CHECK ((status = 'reverted') = (reverted_at IS NOT NULL))
- CONSTRAINT ck_cook_sessions_order CHECK (completed_at >= started_at AND reverted_at >= completed_at AND (reverted_at IS NULL OR completed_at IS NOT NULL))
- CONSTRAINT fk_cook_sessions_meal_plan_id FOREIGN KEY (meal_plan_id) REFERENCES meal_plans (id) ON DELETE CASCADE

1.55. cook_schedule_entries

Bảng 1.55. cook_schedule_entries

Cột	Kiểu và ràng buộc khai báo
cook_session_id	uuid NOT NULL
recipe_id	uuid NOT NULL
step_no	smallint NOT NULL
start_offset_s	integer NOT NULL
end_offset_s	integer NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_cook_schedule_entries PRIMARY KEY (cook_session_id, recipe_id, step_no)
- CONSTRAINT ck_cook_schedule_entries_1 CHECK (start_offset_s < end_offset_s)

- CONSTRAINT ck_cook_schedule_entries_start CHECK (start_offset_s >= 0)
- CONSTRAINT fk_cook_schedule_entries_cook_session_id FOREIGN KEY (cook_session_id) REFERENCES cook_sessions (id) ON DELETE CASCADE
- CONSTRAINT fk_cook_schedule_entries_recipe_id_step_no FOREIGN KEY (recipe_id, step_no) REFERENCES recipe_steps (recipe_id, step_no)

1.56. interactions

Bảng 1.56. interactions

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL
user_id	uuid NOT NULL
recipe_id	uuid NOT NULL
meal_plan_id	uuid
meal_plan_item_id	uuid
type	interaction_type_t NOT NULL
position	smallint
swap_reason	swap_reason_t
ranker_version_id	uuid
propensity	real
occurred_at	timestamptz NOT NULL
received_at	timestamptz NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_interactions PRIMARY KEY (id)
- CONSTRAINT ck_interactions_item_needs_plan CHECK (meal_plan_item_id IS NULL OR meal_plan_id IS NOT NULL)
- CONSTRAINT ck_interactions_position CHECK (position >= 0)
- CONSTRAINT ck_interactions_propensity CHECK (propensity > 0 AND propensity <= 1)
- CONSTRAINT ck_interactions_swap_reason CHECK (swap_reason IS NULL OR type = 'swapped')
- CONSTRAINT fk_interactions_user_id FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE CASCADE
- CONSTRAINT fk_interactions_recipe_id FOREIGN KEY (recipe_id) REFERENCES

recipes (id)

- CONSTRAINT fk_interactions_meal_plan_item_id_meal_plan_id_recipe_id FOREIGN KEY (meal_plan_item_id, meal_plan_id, recipe_id) REFERENCES meal_plan_items (id, meal_plan_id, recipe_id) ON DELETE SET NULL (meal_plan_item_id)
- CONSTRAINT fk_interactions_ranker_version_id FOREIGN KEY (ranker_version_id) REFERENCES model_versions (id)
- CONSTRAINT fk_interactions_meal_plan_id_user_id FOREIGN KEY (meal_plan_id, user_id) REFERENCES meal_plans (id, user_id) ON DELETE CASCADE

1.57. survey_forms

Bảng 1.57. survey_forms

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
version	text NOT NULL
n_anchor	smallint NOT NULL
n_random	smallint NOT NULL
is_open	boolean NOT NULL DEFAULT false
created_at	timestamptz NOT NULL DEFAULT now()

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_survey_forms PRIMARY KEY (id)
- CONSTRAINT uq_survey_forms_version UNIQUE (version)
- CONSTRAINT ck_survey_forms_sizes CHECK (n_anchor >= 0 AND n_random >= 0 AND n_anchor + n_random > 0)

1.58. survey_form_anchors

Bảng 1.58. survey_form_anchors

Cột	Kiểu và ràng buộc khai báo
form_id	uuid NOT NULL

Bảng 1.58 (tiếp theo)

Cột	Kiểu và ràng buộc khai báo
recipe_id	uuid NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_survey_form_anchors PRIMARY KEY (form_id, recipe_id)
- CONSTRAINT fk_survey_form_anchors_form_id FOREIGN KEY (form_id) REFERENCES survey_forms (id)
- CONSTRAINT fk_survey_form_anchors_recipe_id FOREIGN KEY (recipe_id) REFERENCES recipes (id)

1.59. survey_respondents

Bảng 1.59. survey_respondents

Cột	Kiểu và ràng buộc khai báo
id	uuid NOT NULL DEFAULT gen_random_uuid()
form_id	uuid NOT NULL
policy_version_id	uuid NOT NULL
consented_at	timestamptz NOT NULL
completed_at	timestamptz
channel	survey_channel_t NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_survey_respondents PRIMARY KEY (id)
- CONSTRAINT fk_survey_respondents_form_id FOREIGN KEY (form_id) REFERENCES survey_forms (id)
- CONSTRAINT fk_survey_respondents_policy_version_id FOREIGN KEY (policy_version_id) REFERENCES policy_versions (id)

1.60. survey_assignments

Bảng 1.60. survey_assignments

Cột	Kiểu và ràng buộc khai báo
respondent_id	uuid NOT NULL
recipe_id	uuid NOT NULL
display_order	smallint NOT NULL
is_anchor	boolean NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_survey_assignments PRIMARY KEY (respondent_id, recipe_id)
- CONSTRAINT uq_survey_assignments_respondent_id_display_order UNIQUE (respondent_id, display_order)
- CONSTRAINT ck_survey_assignments_display_order CHECK (display_order >= 1)
- CONSTRAINT fk_survey_assignments_respondent_id FOREIGN KEY (respondent_id) REFERENCES survey_respondents (id) ON DELETE CASCADE
- CONSTRAINT fk_survey_assignments_recipe_id FOREIGN KEY (recipe_id) REFERENCES recipes (id)

1.61. survey_ratings

Bảng 1.61. survey_ratings

Cột	Kiểu và ràng buộc khai báo
respondent_id	uuid NOT NULL
recipe_id	uuid NOT NULL
rating	smallint
never_eaten	boolean NOT NULL DEFAULT false
answered_at	timestamptz NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_survey_ratings PRIMARY KEY (respondent_id, recipe_id)
- CONSTRAINT ck_survey_ratings_rating CHECK (rating BETWEEN 1 AND 5)
- CONSTRAINT ck_survey_ratings_1 CHECK ((rating IS NULL) = never_eaten)
- CONSTRAINT fk_survey_ratings_respondent_id_recipe_id FOREIGN KEY (respondent_id, recipe_id) REFERENCES survey_assignments (respondent_id, recipe_id) ON DELETE CASCADE

1.62. survey_favorites

Bảng 1.62. survey_favorites

Cột	Kiểu và ràng buộc khai báo
respondent_id	uuid NOT NULL
recipe_id	uuid NOT NULL

Ràng buộc đặt tên trong CREATE TABLE:

- CONSTRAINT pk_survey_favorites PRIMARY KEY (respondent_id, recipe_id)
- CONSTRAINT fk_survey_favorites_respondent_id FOREIGN KEY (respondent_id) REFERENCES survey_respondents (id) ON DELETE CASCADE
- CONSTRAINT fk_survey_favorites_recipe_id FOREIGN KEY (recipe_id) REFERENCES recipes (id)

1.63. Ràng buộc bổ sung, index và view

Cạnh FK phát sinh từ ALTER TABLE và index không được bỏ qua khi nghiệm thu mô hình. DDL đầy đủ đi kèm trong evidence/schema-source.sql; một số ràng buộc chỉ có ở migration mới.

```
ALTER TABLE ở cuối tệp.
-- Bất biến mà chuẩn hoá dựa vào (docs/sad/04-chuan-hoa.md mục 4) được ép trong CSDL, không
chỉ ở service:
--   * trigger trg_append_only: consent_events, audit_log, pantry_transactions, interactions
chỉ ghi thêm;

ALTER TABLE idempotency_keys ADD CONSTRAINT fk_idempotency_keys_user_id FOREIGN KEY (user_id)
REFERENCES users (id) ON DELETE CASCADE;

ALTER TABLE pantry_lots ADD CONSTRAINT fk_pantry_lots_freshness_model_version_id FOREIGN KEY
(freshness_model_version_id) REFERENCES model_versions (id);

ALTER TABLE pantry_lots ADD CONSTRAINT fk_pantry_lots_source_scan_item_id FOREIGN KEY
(source_scan_item_id) REFERENCES scan_items (id) ON DELETE SET NULL;

ALTER TABLE shopping_items ADD CONSTRAINT fk_shopping_items_source_meal_plan_item_id FOREIGN
KEY (source_meal_plan_item_id) REFERENCES meal_plan_items (id) ON DELETE SET NULL;

CREATE UNIQUE INDEX ux_meal_plan_items_active_slot ON meal_plan_items (meal_plan_id, slot)
WHERE status = 'active';

CREATE UNIQUE INDEX ux_recipes_name_vi_live ON recipes (name_vi) WHERE status <> 'retired';

CREATE UNIQUE INDEX ux_ingredients_name_vi_active ON ingredients (name_vi) WHERE is_active;
```

```
CREATE UNIQUE INDEX ux_vision_class_ingredients_default ON vision_class_ingredients (vision_
class_id) WHERE is_default;

CREATE UNIQUE INDEX ux_cook_sessions_one_live ON cook_sessions (meal_plan_id) WHERE status IN
('in_progress', 'completed');

CREATE INDEX ix_recipe_ingredients_ingredient_id ON recipe_ingredients (ingredient_id);

CREATE INDEX ix_ingredient_allergens_allergen_code ON ingredient_allergens (allergen_code);

CREATE INDEX ix_pantry_lots_user_id_status ON pantry_lots (user_id, status);

CREATE INDEX ix_pantry_transactions_lot_id ON pantry_transactions (lot_id);

CREATE INDEX ix_interactions_user_id_occurred_at ON interactions (user_id, occurred_at DESC);

CREATE INDEX ix_meal_plans_user_id_created_at ON meal_plans (user_id, created_at DESC);

CREATE INDEX ix_meal_plan_items_meal_plan_id ON meal_plan_items (meal_plan_id);

CREATE INDEX ix_scans_user_id_created_at ON scans (user_id, created_at DESC);

CREATE INDEX ix_scan_items_scan_id ON scan_items (scan_id);

CREATE INDEX ix_detections_scan_image_id ON detections (scan_image_id);

CREATE INDEX ix_consent_events_user_id_created_at ON consent_events (user_id, created_at
DESC);

CREATE INDEX ix_dsr_requests_status_received_at ON dsr_requests (status, received_at);

CREATE INDEX ix_refresh_tokens_user_id ON refresh_tokens (user_id);

CREATE INDEX ix_refresh_tokens_family_id ON refresh_tokens (family_id);

CREATE INDEX ix_shopping_items_user_id_status ON shopping_items (user_id, status);

CREATE INDEX ix_audit_log_created_at ON audit_log (created_at);

CREATE INDEX ix_cook_sessions_meal_plan_id ON cook_sessions (meal_plan_id);

CREATE INDEX ix_survey_ratings_recipe_id ON survey_ratings (recipe_id);

CREATE INDEX ix_pantry_transactions_ref ON pantry_transactions (ref_type, ref_id);

CREATE INDEX ix_audit_log_actor_user_id ON audit_log (actor_user_id);

CREATE INDEX ix_dsr_requests_user_id ON dsr_requests (user_id);

CREATE INDEX ix_password_resets_user_id ON password_resets (user_id);

CREATE INDEX ix_detections_scan_item_id ON detections (scan_item_id);
```

```
CREATE INDEX ix_pantry_lots_source_scan_item_id ON pantry_lots (source_scan_item_id);

CREATE INDEX ix_meal_plan_pins_lot_id ON meal_plan_pins (lot_id);

CREATE INDEX ix_shopping_items_source_meal_plan_item_id ON shopping_items (source_meal_plan_
item_id);

CREATE INDEX ix_interactions_meal_plan_item_id ON interactions (meal_plan_item_id);

CREATE INDEX ix_interactions_meal_plan_id_user_id ON interactions (meal_plan_id, user_id);

CREATE INDEX ix_idempotency_keys_created_at ON idempotency_keys (created_at);

CREATE INDEX ix_refresh_tokens_expires_at ON refresh_tokens (expires_at);

CREATE INDEX ix_password_resets_expires_at ON password_resets (expires_at);

CREATE INDEX ix_scans_stuck ON scans (started_at) WHERE status IN ('queued', 'processing');

CREATE INDEX ix_scan_images_purge ON scan_images (scan_id) WHERE purged_at IS NULL AND NOT
contributed;

CREATE VIEW v_current_consents AS
SELECT DISTINCT ON (ce.user_id, pv.purpose_code)
ce.user_id,
pv.purpose_code,
ce.action,
(ce.action = 'grant') AS is_granted,
ce.policy_version_id,
pv.version AS policy_version,
(pv.id = (SELECT p2.id
FROM policy_versions p2
WHERE p2.purpose_code = pv.purpose_code
AND p2.effective_from <= now()
ORDER BY p2.effective_from DESC
LIMIT 1)) AS on_current_policy,
ce.channel,
ce.created_at AS decided_at
FROM consent_events ce
JOIN policy_versions pv ON pv.id = ce.policy_version_id
ORDER BY ce.user_id, pv.purpose_code, ce.created_at DESC, ce.id DESC;

CREATE VIEW v_pantry_lot_balance AS
SELECT l.id AS lot_id,
l.user_id,
l.ingredient_id,
l.status,
l.acquired_at,
GREATEST(0, l.initial_g_min + t.sum_min) AS g_min,
GREATEST(0, l.initial_g_max + t.sum_max) AS g_max,
t.n AS n_transactions
FROM pantry_lots l
CROSS JOIN LATERAL (SELECT COALESCE(sum(x.delta_g_min), 0) AS sum_min,
COALESCE(sum(x.delta_g_max), 0) AS sum_max,
count(*) AS n
FROM pantry_transactions x
WHERE x.lot_id = l.id) t;
```

```
CREATE VIEW v_journal_days AS
SELECT d.user_id,
d.local_date,
d.cooked_sessions,
d.kcal_dishes_per_person,
d.rice_bowls_per_person,
d.kcal_dishes_per_person + COALESCE(d.rice_bowls_per_person, 0) * app_param('rice_bowl_kcal')
AS kcal_per_person,
d.e_target_kcal,
COALESCE(u.urgent_lots_used, 0) AS urgent_lots_used
FROM (SELECT mp.user_id,
(cs.completed_at AT TIME ZONE COALESCE(us.timezone, 'Asia/Ho_Chi_Minh'))::date AS local_date,
count(*) AS cooked_sessions,
sum(k.kcal) AS kcal_dishes_per_person,
sum(mp.rice_bowls_per_person) AS rice_bowls_per_person,
sum(mp.e_target_kcal) AS e_target_kcal
FROM cook_sessions cs
JOIN meal_plans mp ON mp.id = cs.meal_plan_id
LEFT JOIN user_settings us ON us.user_id = mp.user_id
LEFT JOIN LATERAL (SELECT sum(i.kcal_per_person) AS kcal
FROM meal_plan_items i
WHERE i.meal_plan_id = mp.id
AND i.status = 'active') k ON true
WHERE cs.status = 'completed'
GROUP BY 1, 2) d
LEFT JOIN (SELECT mp.user_id,
(cs.completed_at AT TIME ZONE COALESCE(us.timezone, 'Asia/Ho_Chi_Minh'))::date AS local_date,
count(DISTINCT t.lot_id) AS urgent_lots_used
FROM cook_sessions cs
JOIN meal_plans mp ON mp.id = cs.meal_plan_id
LEFT JOIN user_settings us ON us.user_id = mp.user_id
JOIN pantry_transactions t
ON t.ref_type = 'cook_session'
AND t.ref_id = cs.id
AND t.reason = 'cook_deduct'
AND t.urgency_at_event BETWEEN app_param('fresh_u_soon') AND app_param('fresh_u_check')
WHERE cs.status = 'completed'
GROUP BY 1, 2) u
ON u.user_id = d.user_id AND u.local_date = d.local_date;
```